

Grid Risk Command Center

Predicting equipment failures before outages happen

AI / SUSTAINABILITY · TEAM NIRMAAN



Power Failures Strike Without Warning

The Hidden Crisis

Transformer and substation failures cascade into major blackouts affecting thousands. Utilities still rely on calendar-based maintenance—reactive, not predictive.

Early warning signals already exist in real-time sensor data: **temperature, vibration, oil quality, partial discharge, and electrical load**. Weather conditions amplify equipment risk, but operators lack a unified view.

The Gap Operators Face

Operators need to know **what may fail, where, when, and what to do next**—before the lights go out.

- No unified risk view across the grid
- Weather risk data siloed from asset data
- Maintenance scheduled by calendar, not condition
- No actionable prioritization for crews



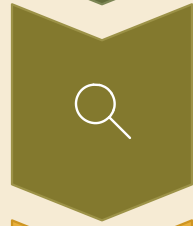
From Raw Telemetry to Operational Decisions

Grid Risk Command Center closes the loop from sensor signal to field action—in a single platform.



Predict

Asset failure probability from sensor patterns



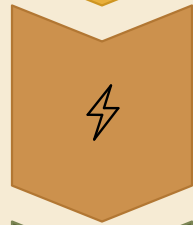
Explain

SHAP-based interpretability surfaces driving risk factors



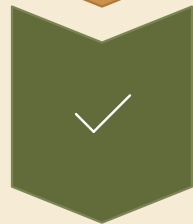
Prioritise

Grid Impact Score ranks what matters most



Simulate

Failures and severe weather scenarios

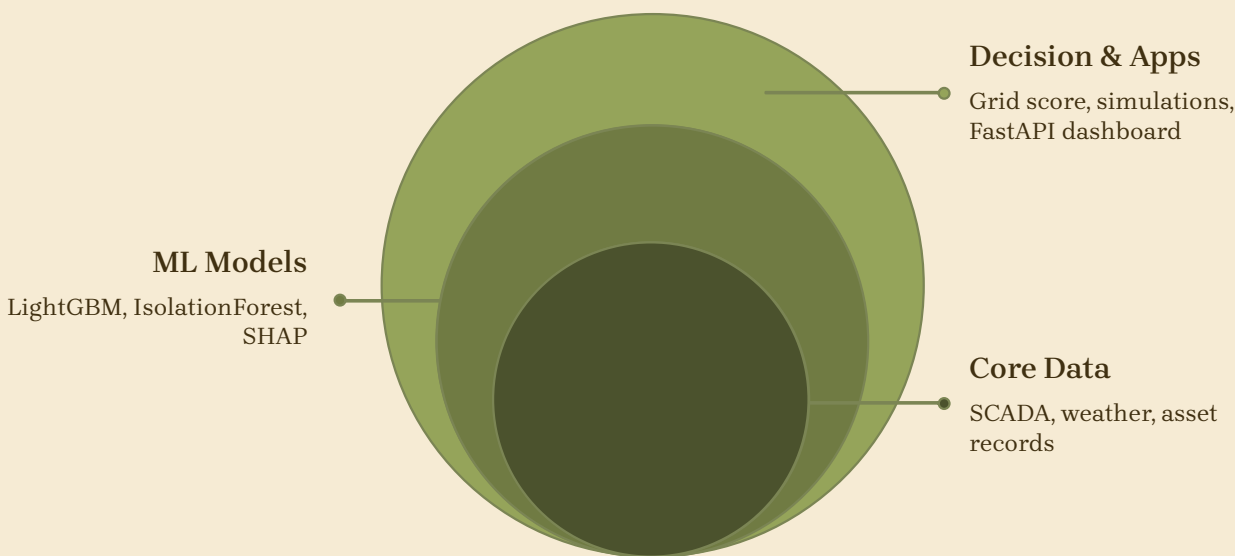


Act

Dispatch, schedule, defer, and log in one interface

System Architecture: Four Layers

Built for Real Operations



Data Layer

Synthetic SCADA telemetry, weather data, asset records, historical incidents

ML Layer

28 engineered asset-health features, LightGBM, IsolationForest anomaly detection, SHAP

Decision Layer

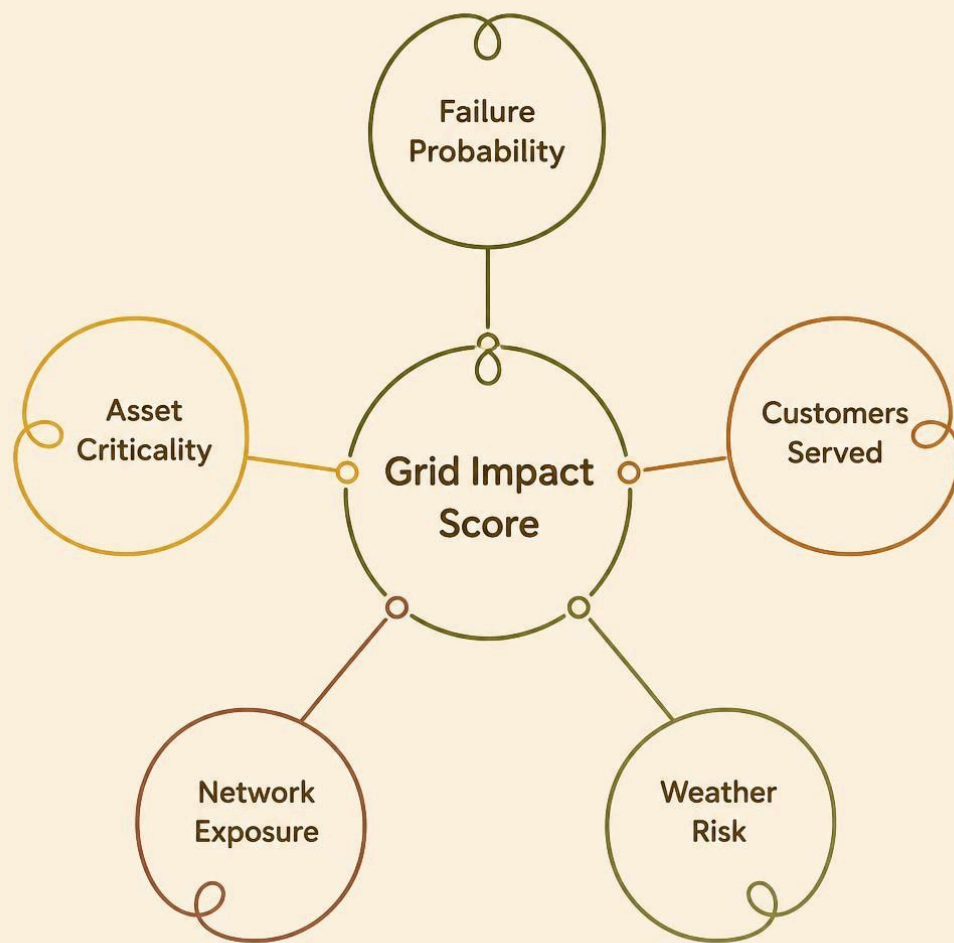
Grid Impact Score, risk prioritisation, crew recommendations, what-if simulations

Application Layer

FastAPI backend, SQLite, JavaScript dashboard with Leaflet maps and Chart.js trends

Grid Impact Score: Ranking What Matters Most

Raw failure probability alone misses the bigger picture. Our composite score combines five critical factors to surface the assets operators should act on *first*.



Why It Changes Everything

Consider Asset **T-1024**: it ranks **#13** by raw failure probability—easy to overlook. Factor in grid impact and it becomes the **#1 maintenance priority**.

- ❏ Prioritising by probability alone leaves the highest-consequence assets unaddressed. Grid Impact Score ensures crews are dispatched where it matters most.

Grounded AI with IBM watsonx.ai

Operations Copilot

IBM watsonx.ai answers operator questions in real-time using the `/ml/v1/text/chat` API with function calling against live backend tools

Grounded Responses

Every answer is grounded in asset telemetry, risk predictions, weather data, maintenance records, and crew availability—with cited tool calls

Always Available

Deterministic local fallback ensures the platform stays operational without API credentials. IBM Cloud IAM secures watsonx.ai authentication



Results: Prediction Meets Prevention

0.97

ROC-AUC Score

On held-out, time-aware test split

28

Engineered Features

Asset-health signals feeding the ML
pipeline

1

Unified Interface

From alert to action without switching
tools

Simulation Capabilities

- Forecast customers affected by a failure scenario
- Identify downstream assets at risk
- Surface nearest available crew
- Estimate outage duration before it occurs

Trust & Compliance Built In

- Every operational action creates a work order and audit record
- Platform uses clearly labelled simulation data—never real customer data
- The goal: **prevent larger outages, not just predict failures**

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Grid Risk Command Center

Utilities act before equipment failure becomes a customer outage.

Predict

AI-powered failure forecasting

Prioritise

Grid Impact Score drives decisions

Prevent

Act before the lights go out

